

・n次式型 $a_{n+1} = p a_n + (n\text{次式})$ $\leftarrow p=1$ なら3階差型

$$1\text{次式} \Rightarrow a_{n+1} + A(n+1) + B = P(a_n + \underbrace{A_n + B}_{1\text{次式}})$$

と変形して②等比型へ

$$2\text{次式} \Rightarrow a_{n+1} + A(n+1)^2 + B(n+1) + C = P(a_n + \underbrace{An^2 + Bn + C}_{2\text{次式}})$$

と変形して②等比型へ

5) A $a_{n+1} = 2a_n + 3n \dots \textcircled{1}$, $a_1 = 1$

$A_{n+1} + A(n+1) + B = 2(A_n + A_n + B)$ ← 華せ白形

$$Q_{n+1} + A_n + A + B = 2Q_n + 2A_n + 2B$$

$$Q_{n+1} = 2Q_n + A_n - A + B$$

①と係数比較して

$A = 3, B = 3$

$$a_{n+1} + 3(n+1) + 3 = 2(a_n + 3n + 3)$$

$$\left(\{a_n + 3n + 3\} \text{ は } \textcircled{27} a_1 + 3 \cdot 1 + 3 \text{ } \textcircled{28} 2 \text{ の等比数列} \right)$$

$$a_n + 3n + 3 = (a_1 + 3 \cdot 1 + 3) \cdot 2^{n-1}$$

$$= 7 \cdot 2^{n-1}$$

$$\therefore a_n = 7 \cdot 2^{n-1} - 3n - 3 //$$

5B $a_{n+1} = 2a_n + n^2 + n + 1 \dots \textcircled{1}$, $a_1 = 1$

$$\boxed{a_{n+1} + A(n+1)^2 + B(n+1) + C} = \boxed{2} \cdot \underbrace{\boxed{a_n + An^2 + Bn + C}}_{\substack{\text{2変式} \\ n \text{ を } n+1 \text{ にした式}}} \leftarrow \text{率せ白形}$$

$$A_{n+1} + An^2 + 2An + A + Bn + B + C = 2An + 2An^2 + 2Bn + 2C$$

$$a_{n+1} = 2a_n + An^2 - 2An + Bn - A - B + C$$

①と係数比較して

$$A=1, B=3, C=5$$

$$a_{n+1} + (n+1)^2 + 3(n+1) + 5 = 2(a_n + n^2 + 3n + 5)$$

($\{a_n + n^2 + 3n + 5\}$ は ㉞ $a_1 + 1^2 + 3 \cdot 1 + 5$ ㉟ 2 の等比数列)

$$a_n + n^2 + 3n + 5 = (a_1 + 1^2 + 3 \cdot 1 + 5) \cdot 2^{n-1}$$

$$= 10 \cdot 2^{n-1}$$

$$= 5 \cdot 2^n$$

$$\therefore a_n = 5 \cdot 2^n - n^2 - 3n - 5 //$$